

****Course Syllabus****

****Artificial Intelligence and Machine Learning****

****Course Code:** CS-301**

****Credits:** 4 (L-T-P: 3-1-0)**

****Semester:** Even Semester 2026**

****Instructor:** Dr. Priya Sharma**

****Email:** priya.sharma@iith.ac.in**

****Office Hours:** Wednesday 2:00 PM - 4:00 PM**

Course Description

This course provides a comprehensive introduction to Artificial Intelligence and Machine Learning. It covers both theoretical foundations and practical implementation of AI/ML algorithms using Python.

Course Objectives

- Understand core concepts of AI and different types of Machine Learning.
- Implement classical AI search algorithms and modern ML models.
- Evaluate model performance and handle real-world datasets.
- Appreciate ethical issues in AI deployment.

Detailed Topics & Weightage

****Unit 1: Introduction to Artificial Intelligence** (15%)**

- History and scope of AI
- Intelligent agents

- Problem solving by searching (BFS, DFS, A*)

****Unit 2: Machine Learning Fundamentals**** (20%)

- Types of learning: Supervised, Unsupervised, Reinforcement
- Data preprocessing, train-test split
- Overfitting, underfitting, bias-variance tradeoff
- Evaluation metrics

****Unit 3: Supervised Learning**** (25%)

- Linear and Logistic Regression
- Decision Trees and Random Forests
- Support Vector Machines (SVM)
- k-Nearest Neighbors

****Unit 4: Neural Networks & Deep Learning**** (20%)

- Perceptrons and Multi-Layer Perceptrons
- Backpropagation
- Introduction to CNNs and RNNs

****Unit 5: Unsupervised Learning & Advanced Topics**** (15%)

- K-Means and Hierarchical Clustering
- Principal Component Analysis (PCA)
- Basics of Reinforcement Learning
- Ethics and bias in AI

****Unit 6: Applications & Case Studies**** (5%)

- Image recognition, NLP, Recommendation systems

Evaluation Scheme

- Mid-Term Exam: ****25%**** — Wednesday, ****6 May 2026****
- End-Term Exam: ****40%**** — Monday, ****15 June 2026****
- Assignments + Quizzes: ****20%****
- Mini Project: ****15%****

Important Deadlines

- Assignment 1: 20 April 2026
- Project Proposal: 5 May 2026
- Final Project Submission: 10 June 2026

Textbooks

1. Stuart Russell & Peter Norvig - Artificial Intelligence: A Modern Approach (4th Edition)
2. Aurélien Géron - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd Edition)

Weekly Schedule (Tentative)

- Weeks 1 - 2: Introduction to AI
- Weeks 3 - 6: ML Basics + Supervised Learning
- Weeks 7 - 9: Neural Networks
- Weeks 10 - 12: Unsupervised Learning
- Weeks 13 - 14: Advanced Topics & Project